Xiaoyu Liu

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Education

Simon Fraser University — Vancouver, BC
Ph.D. in Computing Science — *Jan. 2024 – Present*

- **Relevant Coursework:** Formal Verification, Program Synthesis, Type Theory, Reinforcement Learning

University of Wisconsin–Madison — Madison, WI
Master of Science in Data Science — *Sept. 2020 – May 2022*

Hunan University — Changsha, China
Bachelor of Science in Statistics — *Sept. 2016 – Jun. 2020*

Research Interest

Program Synthesis, Program Testing, Automated Reasoning and Intent Formalization.

Publications

Synthesizing graph queries from demonstrations | To Appear in OOPSLA 2026

- Designed and built DMiner, a **program synthesis** tool that generates Cypher queries from user demonstrations.
- Structured the synthesis process into two stages: first synthesizing graph patterns and generating query sketches, then completing the sketches.
- Created a graph-mining-based algorithm for graph pattern synthesis, using structural constraints to discover relevant patterns and generate query sketches.
- Developed **automated deductive reasoning** rules to identify and prune families of incorrect queries.
- Formulated sketch completion as a **satisfiability-solving and constraint-optimization** problem to generate queries with correct and concise predicates.
- Evaluated DMiner on 90 real-world benchmarks, synthesizing 87 target queries in 0.55 seconds on average and outperforming enumerative and LLM baselines.

Work Experience

Software Engineer — Jan. 2022 – Aug. 2023
DataChat — *Wisconsin, US*

- **Natural Language Understanding Tool**: Fine-tuned **GPT** model to take analytical intent from users, and produce data analysis pipelines consisting of structured tool invocations.
- **Redis TXN Proxy Framework**: Designed a **Redis**-based transactional object store interface. Used Redis native data structures to mitigate nested transaction key lookup from $O(m*n)$ to $O(n)$, and batched read-only query requests to chunk network requests to Redis.
- **API Service**: Implemented API service for multiple web servers with **Go**. Applied a multilayered server-side cache to the data fetching process, reduced the response time by 80%.
- **Data Processing System**: Developed framework to process data with efficient lazy-calculation through **SQL** query chaining functions, as well as a scalable intermediate data objects storage system on **AWS S3**.
- **Engineering Reliability**: Designed the **Python** error hierarchy guideline as a code standard. Improved testing efficiency by implementing automatic testing coverage monitor methods with **Pytest**. Enhanced user-facing error message coverage to 80%, including deciphered database error messages from different dialects.

Technical Skills

Programming/Scripting Languages: Python, Java, Dafny, SQL, Go, C++
Frameworks & Systems: Z3, Docker, Apache Spark, Apache Kafka
Libraries: Hypothesis, CrossHair, SQLAlchemy, PyTorch, TensorFlow
Formal Methods & Automated Reasoning: Program Synthesis, Program Analysis, Deductive Verification, SAT/SMT Solving